

Khulna University of Engineering & Technology
Department of Urban and Regional Planning
BURP 4th Year 2nd Term Regular Examination, 2016
URP 4217
(Coastal and Special Zones Planning)

Full Marks: 210

Time: 3.00 hrs

- N.B. i) Answer any THREE questions from each section in separate scripts.
ii) Figures in the right margin indicate full marks.



Section – A

1. (a) Given the varying nature and extent, how differently Coastal Zones are conceptualized and delineated? Illustrate the typical characteristics of coastal zones. (10)
(b) Discuss the significance of coasts that demands special attention for coastal zone planning. (10)
(c) Critically examine the competing use of the coastal zone/resources and the nature of conflicts. (15)
2. (a) What is meant by ICZM? Differentiate ICZM from conventional coastal zone management. (10)
(b) Discuss the typical steps of an ICZM plan formulation with necessary example. (10)
(c) Classify different types of (coral) reefs. Describe the critical coastal (natural) resources/ecosystems that are subject to degradation due to climatic and non-climatic stressors. (15)
3. (a) How coast and coastal zone in Bangladesh are classified? Illustrate the critical coastal zone management issues in Bangladesh with necessary example. (15)
(b) How do public policies exacerbate the pressure on coastal environment? Illustrate with necessary example. (10)
(c) Discuss the pressure exerted on coastal resources by market forces. How does Wetland Banking help conserve coastal environment and ecosystem? (10)
4. (a) Make an inventory of data and information that are generally required from various organizations/institutions for ICZM planning. (10)
(b) Discuss the key principles and strategies of ICZM planning with their strength and weakness. (15)
(c) Mention some management mechanisms that are required to implement successful ICZM plan. (10)

Section – B

5. (a) Draw a schematic diagram of coastal zone profile. Describe the main elements of the profile with necessary examples. (12)
(b) "Coastal zone management is essential in Bangladesh" - explain the statement with necessary example. (08)
(c) A Category/Class-5 hurricane features winds in excess of 155 mph, although the wind velocity at the center "eye" is very low. Figure A depicts a hurricane hovering over the ocean swell below. The atmospheric pressure 200 mi from the eye is 30.0 in Hg (at point 1, generally normal for the ocean) and the winds are calm. The atmospheric pressure at the eye of the storm is 22.0 in Hg. Estimate the ocean swell at (a) the eye of the hurricane at point 3 and (b) point 2, where the wind velocity is 155 mph. Take the density of seawater and mercury to be 64 lbf/ft³ and 848 lbf/ft³ respectively, and the density of air at normal sea-level temperature and pressure to be 0.076 lbf/ft³. (15)

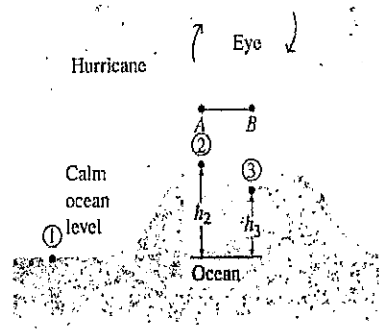


Figure A

6. (a) What do you know about weathering and denudation? (05)
 (b) "Coastal erosion and accretion is complex process involving different factors" – show those factors in a flow diagram. (10)
 (c) How do the wave motion in deep water and wave motion in shallow water differ? Differentiate spring tide from neap tide. (10)
 (d) Determine annual soil (silt loam) loss from an upland farm of 1250 acres in Khulna (Rainfall Erosivity Factor is 300). The slope length is 500 feet with 4 percent slope steepness (topographic factor is 0.762). The soil is covered with matured grass (0.004) and relative erodibilities of silt loam soils is 0.33. (10)
 If slope steepness increases to 5 percent (topographic factor = 1.2), find the changes in soil erosion.
7. (a) Mention the factors that influence sediment yield in a region. (08)
 (b) "Sedimentation is a cycle", do you agree? Justify your position with necessary explanation. (07)
 (c) How is the coastal planning benefited from the study of coastal process (engineering)? Explain with necessary example. (10)
 (d) Waves having a wave period of 10s and wave height of 3.5m break on a quartz sand beach with median grain size ($D_{50} = 0.2 \text{ mm}$). Assume a median grain size D_{50} corresponding to a density of 2650 kg/m^3 , K coefficient = 0.65, $\kappa = 1.0$ and sediment falling velocity = 0.02 m/s for this problem. Calculate volumetric cross-shore sand transport rates in a beach with 5 percent beach slope and water depth of 6m. (10)
8. (a) What are the main challenges of coastal zone management in Bangladesh? (10)
 (b) Discuss the vulnerability of people living in the coastal areas of Bangladesh. (07)
 (c) Suppose, you are working as a Disaster Management Planner at Bangladesh Water Development Board (BWDB). What measures should you adopt to protect Khulna City from coastal flooding? (08)
 (d) Based on the following information, draw the cross-section of the embankment and compute the cross-sectional area of the embankment. (10)
 - Highest flood level (HFL) = 19.70 m
 - Average ground level of river bank = 13.10 m
 - Free board = 90 cm
 - Crest width = 5 m
 - Riverside slope = 1:2 and country side slope = 1:3
 - Setback distance = 30 m.

Khulna University of Engineering & Technology
Department of Urban and Regional Planning
BURP 4th Year 2nd Term Regular Examination, 2016
URP 4273
(Urban Governance)

Full Marks: 210

Time: 3.00 hrs

- N.B.** i) Answer any THREE questions from each section in separate scripts.
ii) Figures in the right margin indicate full marks.

Section – A

1. (a) What is the difference between the concepts of Government and Governance? (08)
(b) What is your understanding regarding the evolution of governance system since 1990s in Bangladesh? (13)
(c) What are the major components of a good governance system? Explain each of those components with appropriate examples. (10)
(d) What are the flaws of the current governance system of Bangladesh? (04)
2. (a) Write a short note on "Cold War". How did Bangladesh's birth as a country was meddled-up in cold war geo-politics? (12)
(b) "Proxies have their own mind" – explain the statement with reference to Cold War experience. (13)
(c) "Cold War officially ended with the fall of Berlin Wall and Neoliberalism officially became a global economic *Mantra*" – do you agree with this statement? Explain your answer. (10)
3. (a) Define social development. Why good governance is in the forefront of global campaign for development and social development. (10)
(b) What are the major campaigns for good governance worldwide? What are the World Bank indicators for good governance? (15)
(c) How do you understand the NGOs as a primary locus of civil society formation in a developing country like Bangladesh? (10)
4. (a) What are the emerging governance issues worldwide? Why do we need a Complex Adaptive System of governance to tackle the emerging issues? (10)
(b) What are four major components of the Complex Adaptive System of governance? Explain those components with a case study and show how they interact and generate different scenarios with a particular focus on the role of government. (18)
(c) What is rule of law? Why do you need a rule of law to ensure good governance? (07)

Section – B

5. (a) "Governance-It goes beyond the Government". Express your thoughts on the statement. (10)
(b) Good governance needs to be responsive, transparent and consensus oriented. Why? (10)
(c) Provide appropriate examples to describe the various levels of governance. (10)
(d) What are the objectives of measuring governance? (05)
6. (a) Which type of assessment is necessary for the following situations? Also mention the reasons behind your choice. (15)
 - i. To find the level of corruption on a massive scale.
 - ii. Applying PRA techniques.
 - iii. Compare data between two or more countries.
 - iv. To compute an input-output matrix for regional economics.
 - v. To apply Geo-Informatics on a particular topic.
- (b) With the help of a spiral diagram show the effect of efficient urban governance to urban development. (10)
- (c) Show the spiral effect of inefficient urban governance with the help of a diagram. (10)
7. (a) Briefly write about the Urban Governance Index (UGI) indicators. (10)
(b) Describe Unobserved Components Model. (15)
(c) Suppose, you are working as a transport planner at Khulna Development Authority. What factors you should consider to design a gender responsive Urban Mass Transport system for Khulna city? (10)
8. (a) What are the content and stages of Citizen Report Cards? (15)
(b) Describe the methodology for community LGSA and Union Parishad LGSA. (20)

Khulna University of Engineering & Technology
Department of Urban and Regional Planning
BURP 4th Year 2nd Term Regular Examination, 2016
URP 4275
(Development Planning)

Full Marks: 210

Time: 3.00 hrs

- N.B.** i) Answer any THREE questions from each section in separate scripts.
ii) Figures in the right margin indicate full marks.

Section – A

1. (a) What is development planning? (05)
(b) Explain with example the criteria for the evaluation of development plans. (10)
(c) Discuss with example the problems of development planning. When is development a curse? Explain. (20)
2. (a) Discuss with example different development plans of Bangladesh. (14)
(b) What is the structure of Planning Commission? (07)
(c) Write down the functions of physical infrastructure division. (07)
(d) Mention the simplified series of steps in the formulation process of ADP. (07)
3. (a) Write down the steps followed in the formulation of the Five Year Plan of Bangladesh. (08)
(b) Evaluate the approaches considered for formulation and implementation of the Sixth FYP. (15)
(c) Discuss with example the areas of improvement considered for higher GDP growth during the Sixth FYP, 2011-15. (12)
4. Write short notes (any five) (5X7=35)
 - a) Real GDP and Nominal GDP.
 - b) Guiding principles of development planning in Bangladesh.
 - c) Vision 2021.
 - d) Development priorities of the perspective plan.
 - e) Articles of state constitution as directives of development planning of Bangladesh.
 - f) Future Orientation Index.
 - g) Happy Planet Index.

Section – B

5. (a) Draw your concept on Third World and Global South. (08)
(b) Discuss the historical evolution of the concept- *Development*. (12)
(c) Illustrate in brief, various dimensions of development. (15)
6. (a) What are the characteristics of Less Developed Countries (LDCs)? (08)
(b) Critically evaluate the WW Rostow's Stages of Growth theory of development. (15)
(c) Discuss the significance of Unbalanced Growth theory in Development. (12)
7. (a) Illustrate Poverty Reduction and Basic Needs paradigm of development. (10)
(b) Discuss the Modernization theory of development. (15)
(c) Illustrate the significance of Neo-liberal theory of development for the development of LDCs. (10)
8. (a) What do you mean by Inclusive Development? (08)
(b) Discuss the Sustainable Development paradigm of Development. (15)
(c) Critically examine the key attributes of Good Development Governance. (12)

Khulna University of Engineering & Technology
Department of Urban and Regional Planning
 BURP 4th Year 2st Term Regular Examination, 2016
URP 4223
(Urban and Regional Economics)

Full Marks: 210

Time: 3.00 hrs

- N.B.** i) Answer any THREE questions from each section in separate scripts.
 ii) Figures in the right margin indicate full marks.

Section – A

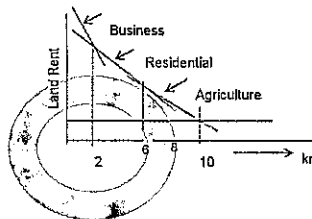
1. (a) 'Trade is a major cause for formation of cities'- explain with examples. (07)
 (b) Suppose a URP, KUET graduate has two job options: (i) stay in Khulna and do freelancing; and (ii) go Dhaka and take up a job in a GIS firm. Considering per month standard living expense 10,000 BDT in Khulna and 15,000 BDT in Dhaka, explain his/her location choice based on the following condition. (25)

	Khulna	Dhaka
Income/month	25,000 BDT	30,000 BDT
One way transport (22 office days/month)	0	120
Social Cost (traffic jam-office days only, pollution, social network loss)	- traffic jam: 0 hrs - pollution: 5 BDT/day - social network loss: 10 BDT/day	- traffic jam: 2 hrs (25 BDT/hr) - pollution: 15 BDT/day - social network loss: 0 BDT/day
Opportunity Cost	100 BDT/day	0 BDT/day

- (i) Explain the ideal location choice of the graduate.
 (ii) In practice, what decision does the graduate make and why?
 (iii) What are the impacts of such practical decision on the urban agglomeration of Dhaka or Khulna?
 (iv) What urban and regional planning policies will you suggest for such a case?
 (c) Why do large cities have higher income/salary? (03)
2. (a) What is demand cone? Explain the Central Place Theory. (08)
 (b) What is bid rent? What is the application or limitation of bid rent in explaining the land use of a city? (05)
 (c) Using the following dataset, draw the bid rent of the city? Explain your understanding about the current land development trend with significance to the urban and regional land use planning, employment and transportation. (22)

	Production per land unit	Sell value per unit	Production cost per unit	Transportation cost per unit
A	50	10	8	0.2
B	45	9	7	0.1
C	40	8	6	0.05
D	45	5	3	0.025
E	30	3	1	0.02

3. (a) What is Zoning? Explain the effects of zoning with reference to land use density, employment and wage. (12)
 (b) Suppose a city permits 100 housing per year and the equilibrium housing price is 10 million per unit in a suburb. If the city authority decides to permit 60 buildings per year, what would be the effect on the housing price? Estimate the price of housing permit and demonstrate the implications of such policy. (18)
 (c) How can you use zoning to promote mixed land use? (05)
4. (a) Assume that bid rent determines land use of a city and services are provided evenly all over the region. Because the city is developing (physically and population) very fast and the government tries to control land development between 6-8 km. Describe the land use scenario after 20 years. (12)



- (b) Suppose a fast growing city has a high housing demand while it allows only 40,000 units per year and the equilibrium price per unit in the inner city area is 10million BDT. Considering the demand, the city authority decides to expand housing supply by 30,000 additional units per year. Graphically explain the impact on the price change and estimate the monetary value of such permission. (15)
 (c) Define urban sprawl. Explain different types of urban sprawl with examples. Explain the application of smart growth as a technique to curb urbanization. (08)

Section – B

5. (a) Why do regions grow? (05)
- (b) Why are location theories important in regional economics? Why does growth of the export sector (basic) drive the economic growth of the region? (10)
- (c) "Bangladesh has no true region except three hill districts but has been delineated into different zones for the sake of usefulness of different works." – explain with example. (10)
- (d) Briefly explain the planning tools what will address the regional growth and development pressures. (10)
6. (a) Mention some units of measure for Economic Growth. Why do you think employment is taken as the popular / common measure? (10)
- (b) "Shift-share analysis breaks down regional employment growths"- explain this statement with necessary examples. (10)
- (c) The following table shows necessary information for industry i of a region for a nation.

	Employment in Industry i	Total Employment	Industry Share of Area
Local area	25	200	?
Local area share of Base Area	?	?	N/A
Base Area	600	12,000	?

- (i) Fill the blanks in the above table. (05)
- (ii) Calculate basic employment in local area of industry i. (05)
- (iii) Calculate location quotient for industry i and decide whether it is self-sufficient, deficit or surplus. (05)
7. (a) Illustrate the circular flow income model with necessary diagrams. (10)
- (b) How does Keynesian regional multiplier relate regional development? Narrate with example. (10)
- (c) The following table shows different industry's employment of a region for a nation along with national growth rate (13.2%), industry sector's national growth rate and regional industry's growth rate.

Industry	Regional Employment (1993)	Industry's National Growth Rate (%)	Regional Industry's Growth Rate (%)
Farm	7,951	-0.1	0.3
Manufacturing	58,516	4.6	4.6
Retail	44,752	13.8	12.6
Finance and Real Estate	16,193	16.5	14.5
Service	62,518	19.3	20.7
All other	63,533	9.7	10.0

- (i) Calculate National Growth share (NS), Regional Growth share (RS) and Industry Mix (IM) separately for each industry. (10)
- (ii) Find out the fast growing and declining industries. (05)
8. (a) Write down the assumptions of the theories of specialization under trade theory. (05)
- (b) Why is Direct Approach used to measure regional GDP having many constraints compared to indirect approach? (05)
- (c) Write down the hypothesis of Von Thünen Model along with his assumptions. (10)
- (d) How does Material Index influence the possible industrial location in Weber's theory? Explain with necessary example. (10)
- (e) Put down the elements of regional economics. (05)

Khulna University of Engineering & Technology
Department of Urban and Regional Planning
BURP 4th Year 2nd Term Regular Examination, 2016

URP 4243
(Hazards and Disaster Management)

Full Marks: 210

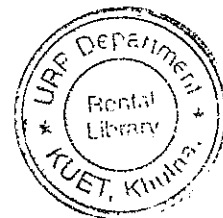
Time: 3.00 hrs

- N.B.** i) Answer any THREE questions from each section in separate scripts.
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Section – A

1. (a) "Disaster risk is a combination of hazard, vulnerability, exposure and capacity" (15)
– explain narrating indicators for each element. Use an example of cyclone here.
- (b) Make a comparative analysis between the following issues. (09)
 - (i) Geological hazards and hydrological hazards
 - (ii) Acceptable risk and residual risk
 - (iii) Direct and indirect economic loss
- (c) Identify and elaborate the risk of top two natural and human-made disasters (11)
considering the vulnerability and exposure of our capital Dhaka.
2. (a) "Vulnerability of any hazard depends on its exposure and sensitivity". Justify (06)
the statement using an example of earthquake in Bangladesh.
- (b) Make a list of issues need to consider in analysing "Flood Hazard" for (10)
Bangladesh.
- (c) Mention the methods used for measuring "Physical Vulnerability" of any (12)
hazard. Elaborate "Matrix Method" of calculating structural vulnerability index.
- (d) "Vulnerability is a multi-dimensional aspect" – differentiate economic, social (07)
and physical vulnerability with appropriate indicators for each.
3. (a) What is your understanding about hazard intensity? Which factors are used to (06)
demarcate the intensity of any hazard? Use an example of flood in Bangladesh.
- (b) How will you differentiate the following terms- (08)
 - (i) Probability and Frequency
 - (ii) Magnitude and Intensity

Use an example of earthquake to clarify the differences.
- (c) What is your understanding about the term "DRR"? Mention some activities (06)
practiced as DRR measures in Bangladesh.
- (d) Provide an example showing the root causes, dynamic pressures and unsafe (15)
conditions of drought vulnerability for Bangladesh.
4. (a) What is your understanding about Community Based Disaster Risk (10)
Management (CBDRM)? Narrate the major steps of conducting CBDRM.
- (b) Specify the steps and procedures of conducting planning stages of Community (15)
Risk Assessment with appropriate example.
- (c) Write short notes on any two of the following issues: (10)
 - (i) Building and training of a community Disaster Risk Management Organization
 - (ii) Community managed implementation
 - (iii) Participatory monitoring and evaluation



Section – B

5. (a) "Risk is a multidisciplinary spatial problem". Briefly explain with necessary example. (05)
- (b) Narrate the immediate and long term impacts of disasters on national development. (10)
- (c) Mention the purposes of disaster management plan. Write down your understanding about the rationales of the counter disaster management plan. (10)
- (d) "Possible links between disaster management planning and development planning should be considered in the planning process of counter disaster management plan". Explain properly with providing necessary examples. (10)
6. (a) Explain the possible mitigation measures for cyclone management in Bangladesh. (10)
- (b) Classify floods based on inundation depth in context of Bangladesh. What are the existing deficiencies in managing flood and what measures can be taken to overcome the deficiencies? (12)
- (c) Write down the flood hazard parameters. Briefly explain the approaches of flood damage assessment. (08)
- (d) "Disaster and national image are inherently interrelated to each other". Write down your rationales to justify the statement. (05)
7. (a) What are the components of Cyclone Master Plan? (05)
- (b) What activities are covered presently under the Cyclone Preparedness Programme (CPP) taken by Government of Bangladesh and the International Federation of Red Cross and Red Crescent Societies (IFRCS)? (10)
- (c) Explain hazards assessment approach with a focus to the use of GIS. (10)
- (d) "Departments/organizations involved in development have no role in day-to-day disaster management activities". Explain with necessary example. (10)
8. (a) "Hazards create risks by exposing pre-existing vulnerabilities". Explain. (05)
- (b) Make a critical analysis about the perquisites of disaster management policy for national development. (10)
- (c) Justify the statement "Prevention is better than cure" in disaster management and planning. (10)
- (d) "Disaster related activities of one phase in disaster management cycle have consequences on other phases". Explain with necessary example. (10)